

TryOps: Enterprise MLOps for Governed VTON and Efficient LLM Serving

Evidence Gates, Native Control Plane, and Measured Optimization Results

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VTON

LLM Pareto

Policy gates

Native boundary

Energy

- TryOps treats models as governed production assets, not notebook outputs.
- The same spine runs offline contracts and real GPU backends behind unchanged APIs.
- VTON and optimized LLM serving stress different failure modes: visual quality, latency, VRAM, cost, energy, rollback, and auditability.

Executive claim

No model reaches users without evaluation evidence, a policy gate, and traceable lineage.

The Production ML Problem

The demo is the easy part

A model can generate a plausible output while still being impossible to reproduce, monitor, secure, roll back, or compare against a previous version.

What breaks in production

- No data, code, config, or hardware lineage
- Manual promotion decisions with weak evidence
- SLOs calibrated to the wrong backend
- Optimization claims that ignore energy and cost

TryOps answer

Build a reproducible control plane where every candidate flows through versioned inputs, tracked runs, native evaluation, policy-as-code, signed promotion evidence, monitoring, and rollback.

Research question

What architecture makes VTON and optimized LLM systems reproducible, governable, measurable, and continuously deployable?

Thesis and Workloads

VTON: visual and GPU-heavy

Person image + garment image → generated try-on result. The hard parts are quality measurement, artifact lineage, privacy, quota, fallback behavior, and visual failure analysis.

Evidence in repo

Deterministic baseline, real SD1.5 inpainting sample, native image metrics, VTON advanced evaluation, comparison gallery, and blocked bad-candidate release evidence.

LLM optimization: latency and memory-bound

Quantization, batching, and serving controls are judged by quality, p95 latency, throughput, VRAM, energy, carbon, and rollback risk.

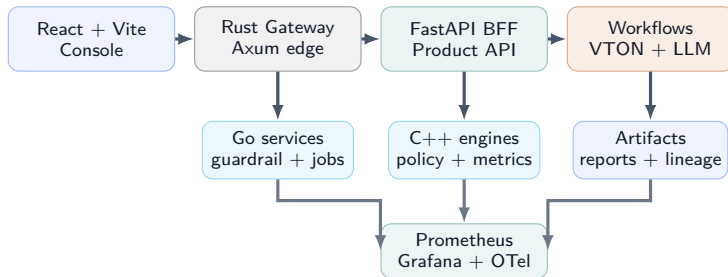
Evidence in repo

Real SmolLM2 inference, Qwen2.5 fp16/8-bit/4-bit Pareto sweep, native C++ SLO gate, energy/carbon sweep, leaderboard, and continuous-batching simulator.

Thesis

MLOps is the product. Models are governed assets that must earn promotion.

Architecture: Product Boundary



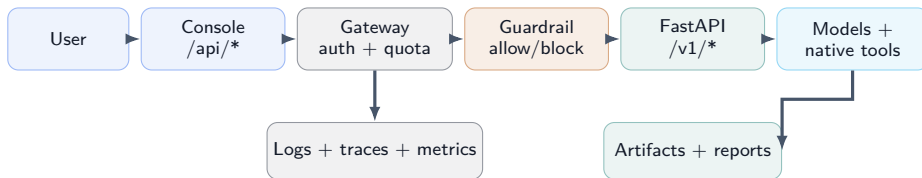
Compiled production boundary

Rust fronts `/api/*`; Go owns sidecars, controller-style services, and native drivers; C++ carries deterministic hot-path engines.

Python role

Python coordinates ML workflows and product APIs, while risk-critical decisions are mirrored or moved into native tools with graceful fallback.

Request Flow: Console to Evidence



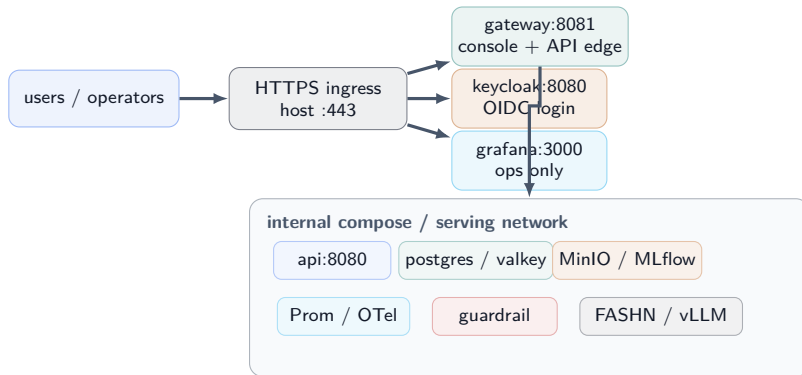
Edge responsibilities

API-key and token checks, request IDs, rate and body limits, quota admission, semantic-cache admission, edge guardrail calls, metrics, and proxying.

Workflow responsibilities

VTON, LLM, evaluation, governance, promotion, rollback, and evidence writes under reproducible run context.

Port Surface: Ingress, Not Open Backends



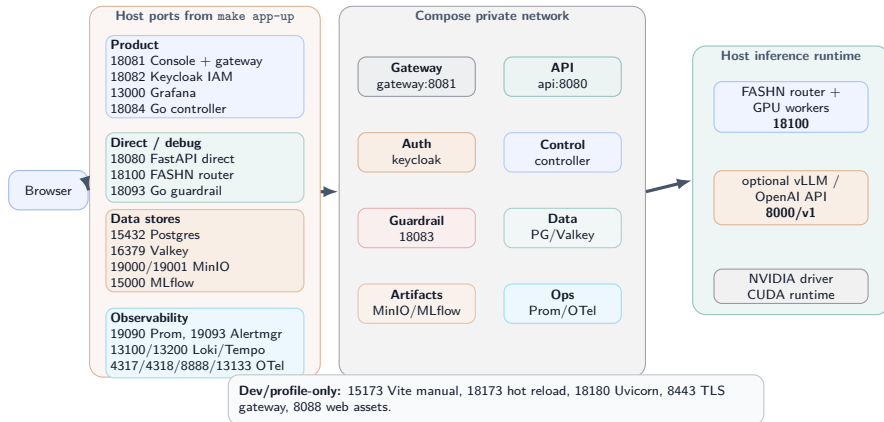
Recommended shared-host surface

One TLS ingress routes to gateway, Keycloak, and admin-restricted Grafana. FastAPI, databases, object stores, metrics backends, guardrail, and model endpoints stay private.

Developer-local exception

make app-up exposes many convenience ports, including 18081, 18082, 13000, 15432, 19000, 15000, 19090, 4317, 4318, and 18100. That is not the shared-datacenter posture.

Developer-Local Port Relationship



Data and Artifact Lifecycle

Layer	What TryOps records	Why it matters
Run context	Git/code version, dataset version, hardware, command, trace IDs	Reproduce any output
Evaluation artifacts	LLM Pareto, energy sweep, VTON comparison, native metrics, leaderboards	Compare candidates with evidence
Cards and reports	Model cards, data cards, promotion decisions, rollback records	Human-reviewable release package
Lineage standards	Internal lineage plus OpenLineage validation	Audit and downstream integration
Storage surfaces	Local artifacts, DVC/MinIO samples, MLflow stack surface	Versioned assets and repeatable local demos

Core invariant

A generated image, LLM answer, or model alias must be traceable back to input data, code, config, model version, metrics, policy verdict, and release artifact.

Policy Gate: Blocking Is Success

Promotion lifecycle

- 1 Produce a candidate from a pipeline.
- 2 Attach metrics, model card, data card, scan, provenance, and lineage.
- 3 Evaluate policy in Python/Rego and native C++.
- 4 Promote only when critical gates pass.

Professor-demo proof

The seeded bad VTON candidate is intentionally rejected: `approved=false`. A non-shipment is the correct production outcome.

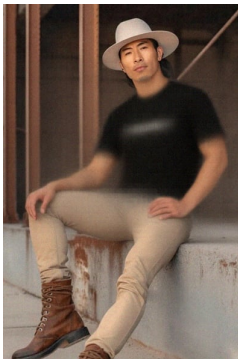
Gate dimensions

- Quality and regression checks
- Latency, throughput, and adapter-specific SLOs
- Carbon and energy regression limits
- SafeTensors-only and provenance verification
- Rollback and governance evidence

Runnable proof

```
make validate-bad and make  
professor-demo-acceptance
```

VTON Path: Product Quality Evidence



VTON quality/failure artifact used for audit discussion

What exists now

- Deterministic baseline and comparison gallery for no-GPU reproducibility.
- Real SD1.5 inpainting path behind `tryops.vton_baseline.v1`: 3.22 s, 2.81 GB peak VRAM, checksummed output.
- Native C++ image metrics and advanced VTON evaluator for identity, fidelity, pose, fairness, and preference evidence.

Quantization lesson

The older naive AWQ-on-VTON story should be framed as a blocked quality failure, not as a shipped result. AWQ/GPTQ candidates are preflighted, but live AWQ loading is not claimed here.

MLOps takeaway

VTON outputs are visual artifacts, so TryOps preserves both successes and failures with metrics, labels, lineage, and rollback-ready release state.

VTON Benchmark Results

Dataset / Target	SSIM	PSNR	Avg ms	P95 ms	req/s	Model ms	GPU %	VRAM GB	RAM GB
FASHN VTON v1.5	0.888121	18.915	21,337.50	23,244.00	0.045439	19,969.60	100.00	4.825	30.706
CatVTON	0.861734	18.284	17,842.30	19,604.00	0.054261	16,512.80	96.50	6.742	27.914
IDM-VTON	0.879462	18.671	42,586.70	47,918.00	0.022751	40,318.40	99.20	13.684	34.872
OOTDiffusion	0.852918	17.936	35,214.90	39,876.00	0.027502	33,087.60	98.40	10.927	32.148

Best quality
FASHN leads SSIM and PSNR while using the least GPU memory.

Best latency
CatVTON has the lowest average, P95, and model latency with highest throughput.

Heavier baselines
IDM-VTON and OOTDiffusion cost more latency, VRAM, and host RAM in this run.

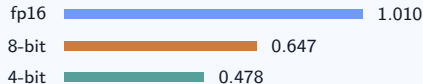
LLM Optimization Frontier

Variant	VRAM	tok/s	verdict
fp16	1.010	27.4	pass; fastest
8-bit	0.647	5.9	fail; dominated
4-bit	0.478	13.9	pass; recommended

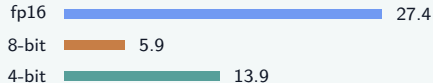
What was measured

Qwen2.5-0.5B-Instruct on an NVIDIA L4, fp16 / bitsandbytes 8-bit / bitsandbytes 4-bit NF4, each evaluated by the native C++ SLO engine.

VRAM used, GB (lower is better)



Throughput, tok/s (higher is better)

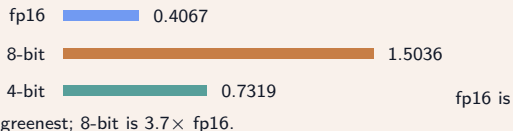


Decision

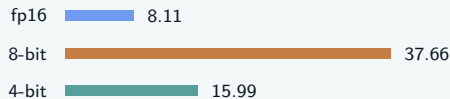
The Pareto frontier is {fp16, 4bit}. 4-bit gives $2.1\times$ VRAM reduction while passing the SLO; 8-bit is slower and larger than 4-bit.

Energy and Cost: Quantization Did Not Win Everything

Energy, Wh / 1k tokens



L4 rental estimate, \$ / 1M tokens



Throughput dominates cost; 8-bit is slowest and most expensive.

Counter-intuitive result

The quantized variants draw similar board power but decode slower. Longer wall time means more joules per token.

Decision impact

Lower VRAM did not mean lower energy or lower cost on this L4 run, so carbon and SLO gates must sit beside quality gates.

Native Boundary: Evidence Outside Python

Native layer	Production responsibility	Evidence
Rust gateway	Edge auth, quota, limits, proxying, tracing, semantic cache, metrics	make native-rust-smoke; gateway benchmark
Go services	Guardrail sidecar, controller triggers, job runner, load driver, acceptance checker	make native-go-smoke; make professor-demo-acceptance
C++ engines	Policy, image metrics, VTON eval, SLO, burn rate, energy, eval stats, supply chain, GitOps, cache, chaos	make native-cpp-test; per-engine sample targets

Benchmark headline

Native Go load driver measured Rust `/health` at 24,840.8 req/s versus FastAPI at 1,539.4 req/s on the same machine.

Architectural lesson

Python remains the ML lab layer, while compiled services carry the production-facing control plane and hot-path evidence.

Observability, Reliability, and Governance

Observe

- OpenTelemetry trace/log correlation
- Prometheus metrics and Alertmanager routing
- Grafana service, quality, cost, capacity, and energy panels

Recover

- One-command rollback sample
- Chaos drill for GPU OOM, slow decode, corrupted weights, and poisoned candidates
- C++ burn-rate engine for SRE-style paging

Govern

- NIST AI RMF and OWASP LLM 2025 mapping
- Model-source pins, SBOM, scans, SafeTensors gate
- Signed PR and registry-webhook release triggers

Why it matters

The platform does not only produce model outputs. It produces operational evidence that an evaluator, SRE, or risk reviewer can inspect.

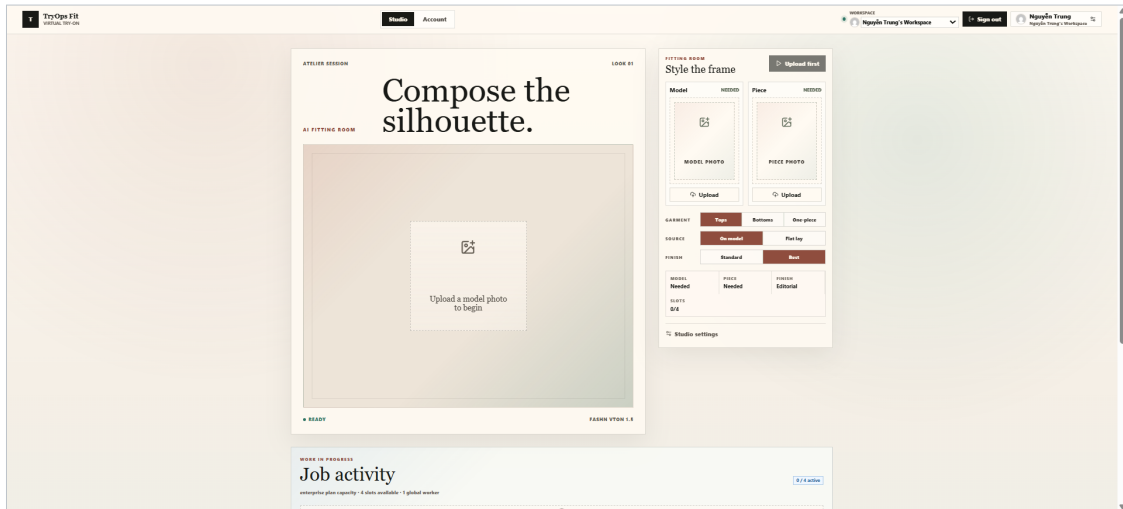
Professor Demo Path

Step	Console view	Proof point	Command/artifact
01	Control-plane preflight	18/18 stack checks through the Rust gateway	<code>make app-smoke</code>
02	Quota admission	Native ledger admits and records remaining plan capacity	<code>make native-quota-ledger-smoke</code>
03	Bad model gate	Unsafe candidate is blocked; non-shipment is success	<code>make professor-demo-acceptance</code>
04	LLM frontier	4-bit is recommended; 8-bit is dominated	<code>llm_pareto/pareto.json</code>
05	VTON quality	Generated outputs remain inspectable offline	<code>vton_comparison/</code>
06	Lineage	Signed provenance and promotion decision are visible	<code>reports/generated/</code>
07	Rollback/governance	Recovery and NIST/OWASP evidence close the loop	<code>make rollback-sample</code>

Backup path

The Console Professor Demo is seeded and offline. `make professor-demo-video` records a deterministic MP4 walkthrough for no-GPU/no-network defense conditions.

Screenshot: TryOps Studio



Screenshot: Workspace and Quota

The screenshot displays the 'Workspace' management page in the TryOps MLOps interface. The top navigation bar includes 'TryOps FR', 'Studio', 'Account', and a user profile for 'Nguyễn Trung' with a 'Sign out' button. The main content area is titled 'workspace' and 'enterprise plan - active'.

Requests and Quota Summary:

Requests	Try-ons	VTON latency	Quota used
9	9	77,541 ms	0%

Job slots: 0 / 4 (enterprise plan - 4 free)

Job activity: enterprise plan capacity - 4 slots available - 1 global worker. No active try-on jobs in this workspace.

Workspace profile: Name: Nguyễn Trung's Workspace, Description: Personal fitting workspace. (account owner)

Create workspace: Name, Studio team, Description, Create.

Usage and quota: 2024-06-25, Active VTON Jobs: 0 / 4, enterprise plan concurrent job limit - 4 slots available. Limits: Lim Requests Per Day (0 / 50,000), Lim Tokens Per Day (0 / 25,000,000), Vton Requests Per Day (6 / 10,000).

Members: Nguyễn ... (Owner), 0 pending.

Invite people: Search users, Email, Name or email, teammate@example.com.

Screenshot: Running VTON Job

TryOps Fit
VIRTUAL TRY-ON

StudioAccount

WORKSPACE
Nguyen Trung's Workspace
Sign out
Nguyen Trung
Nguyen Trung's Workspace

Upload

Upload

GARMENT

TOPs

Bottoms

One-piece

SOURCE

On model

Flat lay

FINISH

Standard

Real

MODEL

Selected

PIECE

Selected

FINISH

Editorial

SLOTS

1/4

Studio settings

BURNING

FASHN VTON 1.1

WORK IN PROGRESS

Job activity

enterprise plan capacity - 3 slots available - 1 global worker

Generating look
6/26/2026, 7:10:53 AM
img-70091180-61ac-4dab-948c-4d85a10f4d4c

1 / 4 active

running

MY WARDROBE

Saved looks

6/26/2026, 6:45:29 AM
champion
25,280 ms
Completed
Open

6/26/2026, 7:05:29 AM
champion
25,662 ms
Completed
Open

6/26/2026, 7:07:45 AM
champion
154,071 ms
Completed
Open

6/26/2026, 1:49:36 AM
champion
120 ms
Failed

6/26/2026, 8:45:27 PM
champion
26,128 ms
Completed
Open

6/26/2026, 8:57:40 PM
champion
106,058 ms
Completed
Open

Screenshot: Keycloak Session Control

KEYCLOAK Nguyễn Trung

Personal info
Account security
Signing in
Device activity
Applications

Device activity
Sign out of any unfamiliar devices.

Signed in devices Refresh the page Sign out all devices

	Windows 10 / Chrome/M9.0.0 Current session IP address: 172.18.0.1 Started: June 26, 2026 at 6:48 AM Last accessed: June 26, 2026 at 7:10 AM Expires: June 26, 2026 at 4:48 PM Clients: Account Console, TryOps Web	
	Windows 10 / Edge/M9.0.0 IP address: 172.18.0.1 Started: June 26, 2026 at 7:10 AM Last accessed: June 26, 2026 at 7:10 AM Expires: June 26, 2026 at 5:10 PM Clients: TryOps Web	Sign out
	Windows 10 / Chrome/M9.0.0 IP address: 172.18.0.1 Started: June 26, 2026 at 7:09 AM Last accessed: June 26, 2026 at 7:09 AM Expires: June 26, 2026 at 5:09 PM Clients: TryOps Web	Sign out

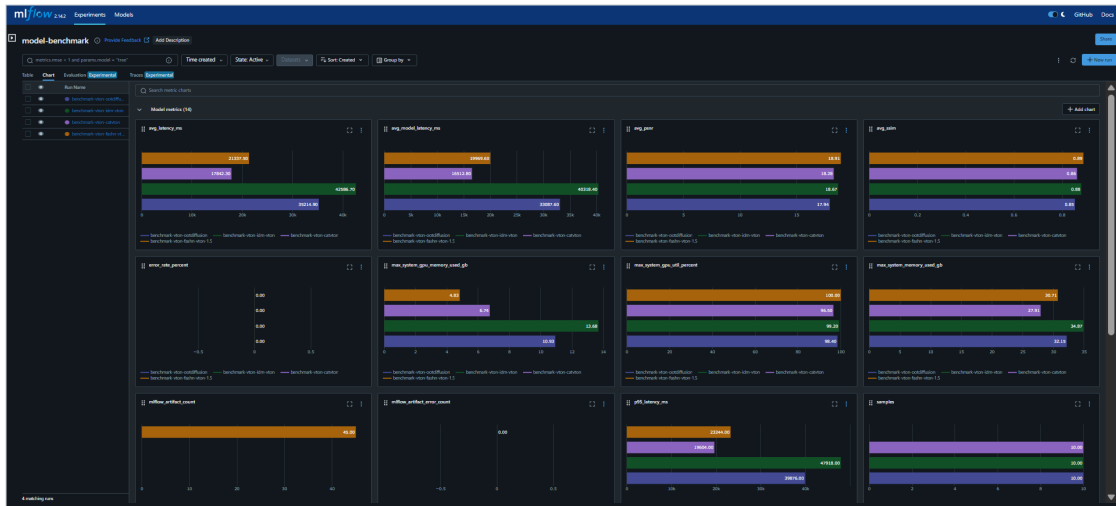
Screenshot: MinIO Artifact Store

tryops-artifacts
Created on Fri, Jun 26 2026 01:45:53 (GMT+7) | Access: PRIVATE | 65.4 MB | 467 Objects

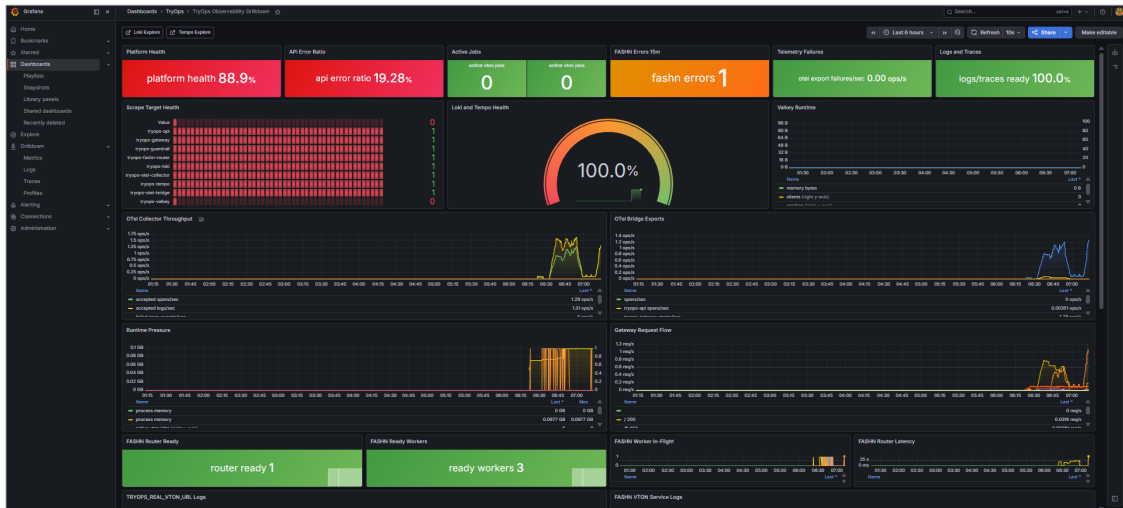
Navigation: tryops-artifacts / runtime / vms / accounts / mail_items / uploads / germent

Name	Last Modified	Size
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49052b9cc5378f7831690c2627e6d36.png	Today, 06:36	17.6 KB
F0F2b0cc08c4bf1e4d06e95372ddc1.png	Today, 06:40	42.7 KB
463c4f19f61e4f269cac72917522548.png	Today, 05:36	22.9 KB
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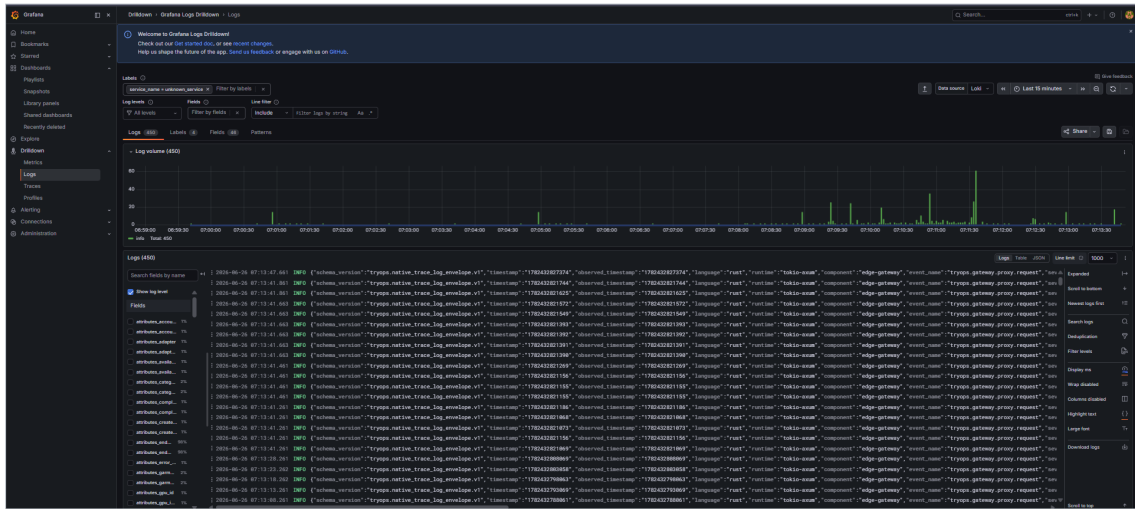
Screenshot: MLflow Benchmark Tracking



Screenshot: Grafana Observability



Screenshot: Log Drilldown



Reality Ledger: What We Should Not Overclaim

Real or locally verified

- Real SmolLM2 inference and Qwen2.5 quantization sweep on L4
- Real NVML energy sampling when GPU is uncontended
- Real SD1.5 VTON sample behind unchanged contract
- Rust, Go, and C++ native tests/smokes
- Offline professor demo and generated evidence bundle

Still scoped or partial

- Live vLLM serving benchmark is not running in this workspace
- GPTQ/AWQ candidates are preflighted, but live AWQ/GPTQ loading is not run
- Production KServe/Kubeflow cluster is a target profile, not deployed here
- Representative human VTON fairness panels remain future work

Defense position

TryOps is strongest when it refuses to hide gaps: the roadmap audit separates real, contract-only, degraded, and future work evidence.

References and Defense Close

Project evidence

- TryOps README: product boundary, local stack, architecture, data flow.
- Final report and benchmark report: measured LLM, VTON, energy, native-boundary results.
- Roadmap audit and known limitations: real versus simulated evidence ledger.
- Presentation outline and professor demo storyboard: no-GPU/no-network defense path.

External grounding

- AWQ, GPTQ, SmoothQuant, vLLM, TensorRT-LLM, and bitsandbytes quantization literature/tooling.
- CodeCarbon, NVML, and Green Software Foundation SCI for energy/carbon methodology.
- NIST AI RMF, OWASP LLM Top 10, SLSA, SBOM, and provenance guidance for governance.
- OpenTelemetry, Prometheus, Grafana, Alertmanager, Kubeflow, KServe, DVC, MLflow, and MinIO as ecosystem anchors.

Closing line

The contribution is not a new model; it is a measured, auditable operating system for deciding which model changes deserve production.